The 8-Pillar Spatial Orchestrator:

A Performative Framework for Real-Time Generative Immersive Environments

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Abstract

This paper presents the 8-Pillar Spatial Orchestrator, a performative framework for real-time generative immersive environments that treats architectural space as a living, score-driven medium. Drawing on R. Murray Schafer's acoustic ecology, Rachel and Stephen Kaplan's Attention Restoration Theory, and contemporary practices in generative art, the system operates through eight independent control parameters — four acoustic (Density, Drift, Spread, Tension) and four visual (Complexity, Coherence, Legibility, Mystery) — organized as a live-performance mixing board. A fifth dimension, Emergent DNA, introduces a biological lifecycle metaphor: visual organisms are born, develop, and dissolve across a non-deterministic lifespan, ensuring no two performances are identical. The framework integrates five generative visual patches (Moon/DNA, Disks, Bloom, Technical, Umbrellas) spatially routed across four projection walls in clockwise or counter-clockwise sweeps. We describe the conceptual basis, technical architecture, and perceptual logic of the system, and position it within the landscape of immersive media art practice. The paper concludes by discussing implications for live performance, therapeutic environments, and future development toward sensor-reactive and AI-adaptive extensions.

Keywords: *immersive environments, generative art, acoustic ecology, soundscape, real-time performance, spatial audio, emergent systems, Kaplan framework, projection mapping, live media art*

1. Introduction

The contemporary landscape of immersive media art is dominated by spectacle. Studios such as teamLab (Tokyo) and Ouchhh (Paris) have demonstrated the cultural appetite for environments in which light, sound, and motion envelop the visitor in continuous, algorithmically generated imagery. Yet for all their visual ambition, most large-scale immersive works operate from pre-rendered or loosely randomized content pipelines — experiences that are generative in

appearance but fixed in their underlying parameters. The audience remains a passive recipient of a predetermined affective arc.

The 8-Pillar Spatial Orchestrator proposes a different model. Rather than pre-composing an experience, it places the artist — or a designated performer — at a live mixing board through which the full perceptual character of the space is modulated in real time. Sound and image are rendered simultaneously from first principles on every frame; nothing is played back from a file. The result is an environment that responds, breathes, and evolves not according to a stored timeline, but according to the live intentions of a human operator working across eight interdependent parameters.

This paper documents the conceptual framework, technical architecture, and aesthetic logic of the system as developed through March 2026. Section 2 situates the work within relevant theoretical traditions. Section 3 describes the 8-Pillar parameter model. Section 4 presents the Emergent DNA lifecycle subsystem. Section 5 documents the five visual patches. Section 6 describes the spatial routing engine. Section 7 discusses the mixing board interface. Section 8 addresses implications and future directions.

2. Theoretical Context

2.1 Acoustic Ecology and the Soundscape

The concept of the soundscape as an artistic and ecological object originates with R. Murray Schafer (1933–2021), the Canadian composer and environmentalist who founded the World Soundscape Project at Simon Fraser University in the late 1960s. In his foundational text The Tuning of the World (1977), Schafer argued that the sonic environment constitutes a form of composition — one that communities inhabit and co-author, whether consciously or not. His taxonomy of soundscape elements (keynotes, soundmarks, and sound signals) provided a vocabulary for analyzing acoustic environments that has since been adopted across disciplines including architecture, urban planning, psychology, and media art.

For the 8-Pillar Orchestrator, Schafer's framework provides more than historical legitimacy. His core claim — that the acoustic environment is a score that can be read, performed, and recomposed — is operationalized directly in the system's Soundscape deck. The four pillars of Density, Drift, Spread, and Tension map onto the primary dimensions by which any soundscape

can be analytically decomposed: how much is happening (Density), where it is going (Drift), how widely it occupies space (Spread), and what emotional valence it carries (Tension).

2.2 The Kaplan Framework and Restorative Environments

The Visualscape parameters draw explicitly from the environmental psychology research of Rachel and Stephen Kaplan, whose Attention Restoration Theory (ART) identifies four properties that predict whether a visual environment will restore directed attention and reduce cognitive fatigue: Being Away, Extent, Fascination, and Compatibility. In their influential formulation, the visual qualities that enable these restorative properties can be parsed as Complexity (information richness), Coherence (organizational clarity), Legibility (inferability of spatial structure), and Mystery (the presence of occluded information that motivates further exploration).

By encoding these four dimensions as live-adjustable parameters, the Orchestrator allows the performer to navigate the full range from cognitively demanding environments (high Complexity, low Coherence) to deeply restorative ones (moderate Complexity, high Coherence, high Legibility, calibrated Mystery) in real time — and to do so in response to perceived audience state, duration of exposure, or dramatic intent.

2.3 Generative and Live Media Art

The system also situates itself within the tradition of generative and live media art, from the early algorithmic works of Harold Cohen and Vera Molnar through to the live cinema practices of artists such as Kurt Hentschlager and Lillevan. In live cinema, the performer-operator occupies a role analogous to a conductor: their gestures shape the unfolding of an audio-visual work in real time, in front of an audience, without the safety net of post-production. The 8-Pillar Orchestrator extends this tradition by foregrounding psychological theory as the compositional grammar — the mixing board is not simply a technical device but a map of human perceptual and affective space.

3. The 8-Pillar Parameter Model

The system is organized around eight independent control parameters, grouped into two functional decks of four. Each parameter is continuously variable across a 0–127 range (matching

MIDI convention), mappable to a physical hardware controller, and output simultaneously to the audio and visual rendering engines.

Table 1: The 8 Pillars — Parameters, Dimensions, and Perceptual Functions

Scape	Pillar	Dimension	Perceptual Function
Soundscape	Density	Temporal	Event saturation per unit time
	Drift	Spatial-temporal	Directional motion of sonic objects
	Spread	Spatial	Field width; mono-to-diffuse continuum
	Tension	Affective	Harmonic dissonance; psychoacoustic arousal
Visualscape	Complexity	Cognitive	Information density of the visual field
	Coherence	Gestalt	Structural order; pattern recognizability
	Legibility	Navigational	Depth inference; spatial comprehension
	Mystery	Motivational	Occluded information; anticipatory engagement

3.1 Soundscape Deck

The Soundscape deck is structured as two independent instrument channels (Deck A and Deck B), each carrying the same four-pillar architecture but with distinct timbral identities. Deck A is characterized by stochastic piano notes and vinyl surface crackle — a textured, intimate sonic world evoking domestic memory and the passage of time. Deck B inhabits a more architectural register: drones whose spatial position is stationary by default but which can be set in motion, circling the room counter-clockwise as the Drift parameter increases.

A crossfader between the two decks allows the performer to blend their timbral characters continuously, while the per-pillar knobs modulate the shared perceptual dimensions independently on each deck. This architecture — borrowed from DJ turntablism — transforms the soundscape from an ambient backdrop into a live compositional instrument.

3.2 Visualscape Deck

The Visualscape deck operates on a single four-parameter channel, though its output is distributed across up to four projection surfaces. Complexity governs the information density of the rendered scene — how many elements are active, how fine their detail, how much the eye must scan to comprehend the whole. Coherence controls the degree to which those elements are organized according to perceivable patterns. Legibility adjusts the clarity of depth and spatial cues. Mystery modulates the degree to which key visual information is occluded, partially revealed, or implied rather than shown.

A secondary crossfader allows transitions between two visual state banks (C and D), enabling smooth morphological movement between, for example, an organic Bloom state and a geometric Technical state.

4. Emergent DNA: A Biological Lifecycle for Generative Organisms

Perhaps the most conceptually distinctive element of the system is the Emergent DNA subsystem, which governs the lifecycle of visual organisms within the rendered scenes. Rather than rendering static or perpetually cycling patterns, the system spawns visual entities that pass through identifiable biological phases: birth, growth, maturity, senescence, and dissolution.

Each organism carries a unique generative identity — a DNA signature — that determines its morphological characteristics: the rate of its spiral expansion, the curvature of its branching structures, the rhythm of its pulsation. This signature is randomly seeded at spawn time and is not repeated across sessions. The organism lives for a finite duration, drawn down by an entropy parameter that can be adjusted by the performer, and when its lifecycle is complete it dissolves back into the visual field — leaving, in some scenes, traces or residue that influence the emergence of the next organism.

This biological metaphor serves several functions simultaneously. Perceptually, it introduces an arc of anticipation and loss into the viewing experience: audiences become invested in individual organisms, notice their aging, and register their dissolution as an event. Philosophically, it enacts a claim about the relationship between life and time in generative systems — rejecting the infinite repeatability of digital media in favor of singular, mortal events.

Technically, it ensures that the system is never in the same state twice, even within a single performance, making the experience genuinely non-reproducible.

The lifecycle is expressed through four observable phases: **Alive** (full vitality, maximum morphological activity), **Developing** (growth phase, structures extending and elaborating), **Dying** (entropy increasing, forms beginning to dissolve), and **Dissolved** (organism absent; field reset for next spawn). The performer can intervene at any phase with Spawn (restart) or Kill (immediate dissolution) commands.

5. Visual Patches

Five visual patches constitute the current scene vocabulary of the system. Each patch is a fully procedural, real-time generative program whose visual output is driven by the eight pillar parameters and the DNA lifecycle state. Scenes are selected via a patch bank interface and can be transitioned between using fade envelopes.

Table 2: Visual Patch Inventory

ID	Patch Name	Behavior	Perceptual Intent
A	Moon u/DNA	Stationary; Wall 1 anchor	Contemplative presence; organism at rest
B	Disks	Concentric rings; counter-clockwise drift	Hypnotic rotation; vestibular engagement
C	Bloom	Organic spirals; fade-to-bloom transition	Biological emergence; birth metaphor
D	Technical	Geometric; fade-to-black then re-emerge	Structural inquiry; cognitive mapping
E	Umbrellas	Concave bowl forms; counter rotation	Protection; shelter; overhead canopy affect

5.1 Moon u/DNA (Patch A)

The foundational patch of the system. A single large luminous form — evoking a moon, an egg, or a cell membrane — floats anchored to Wall 1. Its interior carries the Emergent DNA structure: a branching, slowly animating lattice that represents the organism's internal state. At low Complexity, the Moon reads as simple and contemplative; as Complexity increases, its internal

structure becomes increasingly elaborate. At high Mystery, portions of the internal lattice are occluded by translucent gradients, making the organism's nature inferential rather than explicit.

5.2 Disks (Patch B)

Concentric ring systems whose rotation is governed directly by the Drift parameter. At zero Drift, the disks are stationary. As Drift increases toward maximum, the rings separate into independent orbital layers, each circling the center at a rate proportional to its radius, producing a differential rotation effect that creates strong illusory depth. The counter-clockwise direction is the default; the spatial routing engine can mirror the direction per-wall, creating opposing rotational fields across adjacent surfaces.

5.3 Bloom / Four Walls (Patch C)

The most biologically explicit patch. Organic spiral forms emerge from seed points distributed across the projection surface, expanding outward in patterns governed by phyllotactic mathematics — the same angular relationships that govern the arrangement of seeds in a sunflower or leaves on a stem. The Bloom patch is tightly integrated with the Emergent DNA system; each bloom event corresponds to a DNA organism spawn, and the organism's unique signature shapes the spiral's expansion rate, curvature, and color temperature.

The Bloom patch is designed for four-wall projection, with bloom events seeded independently on each wall but sharing a common DNA pool. The Spread parameter governs how far bloom events migrate across wall boundaries, creating a sense of organic overflow between surfaces.

5.4 Technical (Patch D)

A geometric, architectural patch that provides formal contrast to the organic vocabulary of Bloom. Rectangular and angular forms — reminiscent of architectural drawings, circuit diagrams, or urban grid maps — organize and reorganize themselves according to the Coherence and Legibility parameters. At maximum Coherence, the Technical patch reads as orderly and rational; as Coherence decreases, the forms begin to misalign, overlap, and generate interference patterns. The patch supports a Fade to Black transition, making it suitable for moments of dramatic punctuation or reset within a performance.

5.5 Umbrellas (Patch E)

A patch of concave, bowl-shaped forms distributed across the field, oriented to suggest overhead canopies or protective shelters. The Umbrellas patch engages the affective dimension of shelter and enclosure, providing the highest available Mystery score of the five patches — much of each form's internal structure is implied rather than shown. The patch moves in counter-clockwise rotation as Drift increases, and responds strongly to the Tension parameter, which causes the bowl forms to flatten, invert, and ultimately revert through transitional geometries.

6. Spatial Routing Engine

The four-wall architecture of the installation requires a spatial routing layer that governs how visual and sonic content is distributed, synchronized, and swept across projection surfaces. The routing engine operates on two levels: per-wall assignment (which patch or mix of patches is rendered on each wall) and sweep animation (continuous spatial motion of content across walls).

The Wall Sweep fader controls the propagation of the active scene state across walls in clockwise sequence. At zero, only Wall 1 carries the active scene; as the fader increases toward maximum, the scene state propagates to Wall 2, then 3, then 4 in sequence. This creates a sense of spatial expansion — the environment grows as the fader opens. In the full-sweep position, all four walls share the active state.

The Direction control governs whether content movement and sweep progression is clockwise or counter-clockwise. This parameter interacts with the Drift pillar: when Drift is non-zero, sonic and visual elements move around the room in the direction specified by the Direction control. The All Counter parameter provides an override that sets all patch motions to counter-clockwise simultaneously, creating a unified rotational field.

The Return to A function serves as a system reset: it immediately transitions all parameters toward their Patch A default state, re-anchoring the Moon/DNA form on Wall 1 and resetting the spatial routing to its initial configuration. This function is intended for use at the conclusion of a performance cycle or as a structural reset during a long-duration installation.

7. The Mixing Board Interface

The performer interface is designed as a dual-deck mixing board, drawing directly on the visual and tactile conventions of DJ and live audio performance. This choice is not merely aesthetic: the mixing board metaphor frames the artist as a live conductor of perceptual space rather than a programmer or technician. The interface is intended to be performative in itself — legible to an audience as an act of real-time composition.

The interface comprises the following control surfaces:

- Two sound decks (A and B), each with four rotary knobs for Density, Drift, Spread, and Tension
- A vertical A-to-B crossfader for timbral blending between decks
- A horizontal sound crossfader for additional blend control
- One visual parameter deck with four rotary knobs for Complexity, Coherence, Legibility, and Mystery
- A C-to-D visual crossfader for morphological transitions between scene states
- A five-button scene bank for patch selection (A through E)
- A four-stage Wall Sweep fader with visual indicators
- A Direction control (Counter-clockwise to Clockwise)
- A Return to A reset trigger
- The Emergent DNA lifecycle monitor with Spawn and Kill controls

The software implementation renders this interface as an interactive HTML/JavaScript application capable of running in any modern web browser, with all controls mapped to MIDI CC messages for connection to physical hardware controllers such as the Korg nanoKONTROL or Behringer X-TOUCH series. The separation of interface from rendering engine allows the performer's display and the projection surfaces to operate on independent machines connected via a local network.

8. Discussion and Future Directions

8.1 The Performer as Perceptual Conductor

What distinguishes the 8-Pillar Orchestrator from related generative systems is the centering of live human judgment as the primary compositional force. The system does not automate the relationship between psychological intention and perceptual output; it makes that relationship explicit and requires the performer to navigate it in real time, in front of an audience, in response to the observed state of the room. This is a fundamentally different model from ambient generative installation, where an algorithm manages the environment autonomously, and from interactive art, where the audience drives the parameters. Here, the performer occupies a role that is closer to a conductor or a live sound engineer: technically skilled, perceptually attuned, and publicly accountable.

8.2 Therapeutic and Institutional Applications

The Kaplan framework's origins in restorative environment research suggest a natural application domain beyond the gallery: clinical and therapeutic spaces. Waiting rooms, infusion centers, palliative care environments, and rehabilitation facilities are increasingly recognized as spaces where visual and acoustic design has measurable effects on patient outcomes. A performative system capable of real-time adjustment of restorative parameters — calibrated to the needs of a specific population on a specific day — represents a significant departure from the static design solutions currently employed in these contexts. Future work should include empirical studies measuring the psychophysiological effects of performer-modulated environments on clinical populations.

8.3 Sensor-Reactive Extensions

The current system is entirely performer-driven. A natural extension would introduce environmental sensors — acoustic level meters, infrared occupancy sensors, physiological wristbands — that feed real-time data back to the mixing board as either performance aids or autonomous modulators. In the performance aid model, sensor data informs the performer's decisions without directly driving parameters; in the autonomous model, sensor data directly adjusts parameters within performer-set ranges. Both models raise interesting questions about agency, authorship, and the boundary between composition and improvisation.

8.4 AI-Adaptive Parameter Control

A further extension would introduce machine learning models trained on audience response data to recommend or automate parameter adjustments toward specified perceptual goals. A system trained on physiological and behavioral correlates of restorative attention, for example, could act as a co-pilot to the performer — suggesting adjustments when the environment drifts outside a target range. This raises significant questions about the nature of authorship in generative performance that deserve sustained critical attention.

8.5 Limitations

The current system has not yet been evaluated through formal audience studies. The perceptual claims made in this paper — regarding the mapping between the eight parameters and their intended psychological effects — are grounded in theoretical frameworks but require empirical validation. Future iterations of the work will include structured audience response protocols and, where possible, physiological measurement. Additionally, the current five-patch visual vocabulary, while coherent, represents a limited subset of the system's potential scene space; expansion of the patch library remains an ongoing priority.

9. Conclusion

The 8-Pillar Spatial Orchestrator represents a systematic attempt to translate two foundational bodies of environmental psychology research — Schafer's acoustic ecology and the Kaplan restorative environment framework — into a live performance instrument for immersive media. By structuring the full perceptual character of an architectural space across eight independently variable parameters, and by introducing a biological lifecycle metaphor through the Emergent DNA subsystem, the system enables a form of real-time perceptual composition that has no direct equivalent in existing immersive media practice.

The work argues, implicitly, that the design of experiential environments is a performative act — one that deserves the same rigor, craft, and live accountability that we accord to musical performance. A room is a score. The body that inhabits it is both instrument and audience. The Orchestrator gives the artist a way to conduct that score, in real time, with intention, and without recourse to the fixed and the predetermined.

The practical implications extend well beyond the gallery: therapeutic environments, educational spaces, public installations, and live performance venues all stand to benefit from a

framework that treats the acoustic and visual dimensions of space as a unified, continuously adjustable medium. The work documented here represents an early articulation of that framework; its further development, empirical validation, and extension into sensor-reactive and AI-adaptive domains will be the subject of continuing research.

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Appendix: Recommended Journals for Submission

The following journals are recommended as target venues for this work, in approximate order of fit:

1. Leonardo (MIT Press / Leonardo/ISAST)

The flagship peer-reviewed journal at the intersection of art, science, and technology. Publishes both research papers and artist statements. Strongly aligned with the hybrid theoretical-practice nature of this work. Impact Factor: ~1.2. Submission: leonardo.info/isast/journal.

2. Organised Sound (Cambridge University Press)

A peer-reviewed journal focused on electroacoustic and computer music, with strong coverage of soundscape composition and spatial audio. The Soundscape pillar framework maps directly onto this journal's core scope. Impact Factor: ~0.7.

3. Journal of Environmental Psychology (Elsevier)

The primary venue for Attention Restoration Theory research. A submission to this journal would need to foreground the Kaplan framework empirical claims and ideally include audience response data. High prestige; competitive. Impact Factor: ~5.8.

4. Digital Creativity (Taylor & Francis)

A peer-reviewed journal covering creative uses of digital technology across art, design, music, and performance. Well-suited to the practice-based framing of this work. Impact Factor: ~1.8.

5. ACM NIME (New Interfaces for Musical Expression) Proceedings

The premier conference/proceedings venue for novel instrument and interface design in music and media. The mixing board interface architecture and MIDI routing logic are directly in scope. Conference proceedings rather than journal; strong community, rapid dissemination.

6. ACM CHI / ACM Creativity and Cognition

For submissions emphasizing the human-computer interaction dimensions of the mixing board interface and the performer's real-time cognitive model. Creativity and Cognition is particularly well-suited for the Emergent DNA lifecycle framing.

7. Presence: Teleoperators and Virtual Environments (MIT Press)

A journal focused on immersive and virtual environment experience. Relevant if audience perceptual presence data is incorporated. Strong fit for the spatial routing and four-wall immersion architecture.